Chapter 6: Trigonometric Functions — Right Triangle Approach

Name: ____________________ Date: ____________________

Instructions: Answer all questions. Show complete work and justify your steps. Give *exact* values (in terms of π and radicals) unless otherwise specified. Clearly indicate whether angles are in degrees or radians. For applications, include appropriate units.

Each problem begins on a new page.

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1. Angle measure and arc/sector computations. (12)
   (a) Convert $\dfrac{5\pi}{3}$ radians to degrees.
   (b) Convert $-420°$ to radians and give a coterminal angle in $[0, 2\pi)$.
   (c) On a circle of radius 7 units, find the arc length and the area of the sector cut off by a central angle of 2.4 radians. (Give decimal answers.)

2. In $\triangle PQR$ with $\angle PQR = 90^\circ$, suppose $\angle QPR = 38^\circ$ and $QR = 12$. Find PR and PQ to the nearest hundredth. (10)

3. Find the measure (to the nearest degree) of the acute angle formed by the x-axis and the line $2x + 3y = 7$. Explain briefly. (10)

4. Evaluate each exactly, giving answers in radians. (10)

   (a) $\arcsin\left(-\dfrac{\sqrt{2}}{2}\right)$

   (b) $\arccos\left(-\dfrac{\sqrt{2}}{2}\right)$

   (c) $\arctan(-1)$

5. Let $y = \arctan(2x)$. Express $\sin y$ and $\cos y$ entirely in terms of x (no trigonometric functions remaining). State any domain restrictions used. (14)

6. Solve $\triangle ABC$ given $A = 49°$, $B = 63°$, and $a = 12$. Find b, c, and C (nearest tenth where appropriate). Indicate which law you used. (12)

7. Given $A = 27^\circ$, $a = 9$, and $b = 13$, determine the number of possible triangles. Solve all that exist (find all missing sides and angles, to the nearest tenth of a unit and degree). (12)

8. Two airplanes take off from the same airport at the same time. One flies due south at 300 miles per hour. The other flies $30°$ east of due north at 240 miles per hour. To the nearest mile, how far apart are the planes after 2 hours? (12)

9. In a triangle with side lengths 7, 9, and 15, determine whether the angle opposite the longest side is acute, right, or obtuse. Then find its measure to the nearest degree. (10)

10. Find the area of a triangle with side lengths 5, 7, and 9 using Heron's formula. Give an exact expression and a decimal approximation. (10)